BASHAMATE SYSTEM • TEST DOCUMENTATION

Software Test Case Specification

Component: Review Management • Class: `Review` • Target Method: `submitReview()`

Field	Details
Test Case ID	**TC-BM-REV-01**
Test Scenario	Verify that the system blocks review submission and displays an error message when rating value exceeds the 5-star maximum boundary limit (Boundary Value Analysis).
Preconditions	1. Student account is active, authenticated, and logged into BashaMate platform. 2. Student has an existing booking or rental interaction with target Listing ID `LIST-501`. 3. The "Write a Review" submission form modal is open and responsive.
Test Steps	1. Navigate to the target Listing details page for `LIST-501`. 2. Click on the **Write a Review** button. 3. Enter rating value **6** in the numeric input field (Permitted boundary: 1 to 5). 4. Enter review feedback: "Convenient location and very hospitable host." 5. Click on the **Submit Review** button (triggers `submitReview()`).
Input Data	• **reviewerId:** "STU-1024" • **targetId:** "LIST-501" • **rating:** 6 *(Boundary test value: max + 1)* • **comment:** "Convenient location and very hospitable host."
Expected Result	System blocks submission, prevents database write, and triggers error alert: *"Submission failed: Rating must be an integer between 1 and 5."* No new review record is inserted.
Actual Result	(To be completed during test execution)
Status	(Pass / Fail)
Remarks	Applies Boundary Value Analysis (BVA) on the upper limit of the 1–5 star rating scale (5+1 boundary test).

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Software Test Case Specification

Component: Review Management • Class: `Review` • Target Method: `editReview()`

Field	Details
Test Case ID	**TC-BM-REV-02**
Test Scenario	Verify that an authenticated author can successfully update the rating score and text commentary of an existing posted review (Functional Testing).
Preconditions	1. Student `STU-1024` is authenticated and logged into the BashaMate system. 2. An existing review record `REV-201` authored by `STU-1024` exists in the database with initial rating 3 and comment *"Average room condition."*
Test Steps	1. Navigate to the **My Reviews** section under Student Dashboard. 2. Locate review record `REV-201` and click **Edit Review**. 3. Change the rating field from **3** to **5**. 4. Update the comment field to: "Renovated recently; significantly improved facilities and cleanliness." 5. Click on the **Save Changes** button (triggers `editReview()`).
Input Data	• **reviewId:** "REV-201" • **reviewerId:** "STU-1024" • **updatedRating:** 5 • **updatedComment:** "Renovated recently; significantly improved facilities and cleanliness."
Expected Result	System accepts valid updates, persists new values to the database, updates modification timestamp, and displays confirmation: *"Review updated successfully."* Listing view reflects the 5-star rating.
Actual Result	(To be completed during test execution)
Status	(Pass / Fail)
Remarks	Validates data update integrity, authorization constraints, and instantaneous UI re-rendering.

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Software Test Case Specification

Component: Review Management • Class: `Review` • Target Method: `calculateRating()`

Field	Details
Test Case ID	**TC-BM-REV-03**
Test Scenario	Verify that the system accurately calculates and updates the arithmetic mean rating score for a listing/landlord when a new rating is processed (Calculation & Aggregation Testing).
Preconditions	1. Target listing `LIST-501` has exactly two existing confirmed reviews with rating scores: `[4, 5]` (Current average: 4.50). 2. A new third review with rating score 3 is verified and submitted.
Test Steps	1. Submit a new review with rating **3** for listing `LIST-501`. 2. Trigger internal execution of `calculateRating()` upon review insertion. 3. Query the updated cumulative rating attribute on `LIST-501` in the database. 4. Open the public listing page for `LIST-501` and inspect the displayed rating banner.
Input Data	• **Target:** "LIST-501" • **Existing Ratings:** [4, 5] (Sum = 9, Count = 2) • **New Rating:** 3 • **Formula:** (4 + 5 + 3) / 3 = 12 / 3 = 4.00
Expected Result	System calculates exact floating-point average of **4.00**, updates `Float rating` in `Listing` and `Landlord` entities, and displays "4.0 ★ (3 Reviews)" on the UI.
Actual Result	(To be completed during test execution)
Status	(Pass / Fail)
Remarks	Validates mathematical accuracy, float precision formatting (2 decimal places), and backend data synchronization.